

ML - Assignment

Hope Artificial Intelligence

Scenario Based Learning

Scenario Given:

A company works with number of employees, all the works are dependents on the employees. Even if one of the employees resign the job immediately then assigned work will be not finished at the time, so delivery of the project to the clients will be delayed. Company planned to make solution for this, they want to know which employee may resign next. If they know previously, they can arrange alternative to avoid such problem. As an AI Engineer you must give Solution to this.

A) How will you achieve this in AI?

As the dataset is populated with Numerical values as key columns for training we consider the Machine Learning Approach, and the Dataset Labeled column "Status" is a text based bivariate "Active" and "Inactive" value which can be embedded as 1 and 0 for Active and Inactive status.

B) Find out the 3 -Stage of Problem Identification

As discussed earlier Stage 1 - "*Machining Learning*" Approach ,Stage -2 As the provided dataset shows the outcome/result as status column it is a "*Supervised*" Learning type and Stage -3 as the outcome of Status column is bivariate ,so it is a "*Classification*"

C) Name the project

As in today's Technology Industry Skilled Workforce are the Real Assets of an Organisation , Controlling the Attrition Strategically with tools , Such a tool should be named as "***Attrition Detection Tool***".

D) Create the dummy Dataset.

A Sample of Dummy Dataset with only 13 created with columns with a output Column like

- | | |
|--------------------------------------|-------------------------------------|
| 1.Employee Name | 2.Employee ID |
| 3.Job Satisfaction | 4.Salary Growth / Increment History |
| 5.Years Since Last Promotion | 6.Overtime Hours |
| 7.Attendance / Absenteeism | 8.Manager Relationship Score |
| 9.Grievance Count | 10.Work-Life Balance Score |
| 11.Tenure in Current Role | 12.Team Attrition Rate |
| 13.Status (Labeled Data/Output Data) | |

Please find the file "Employee_Attrition_Dummy_Data.csv" CSV dataset which is a portable and widely adopted format along with this document.

